11 AI and Ethics

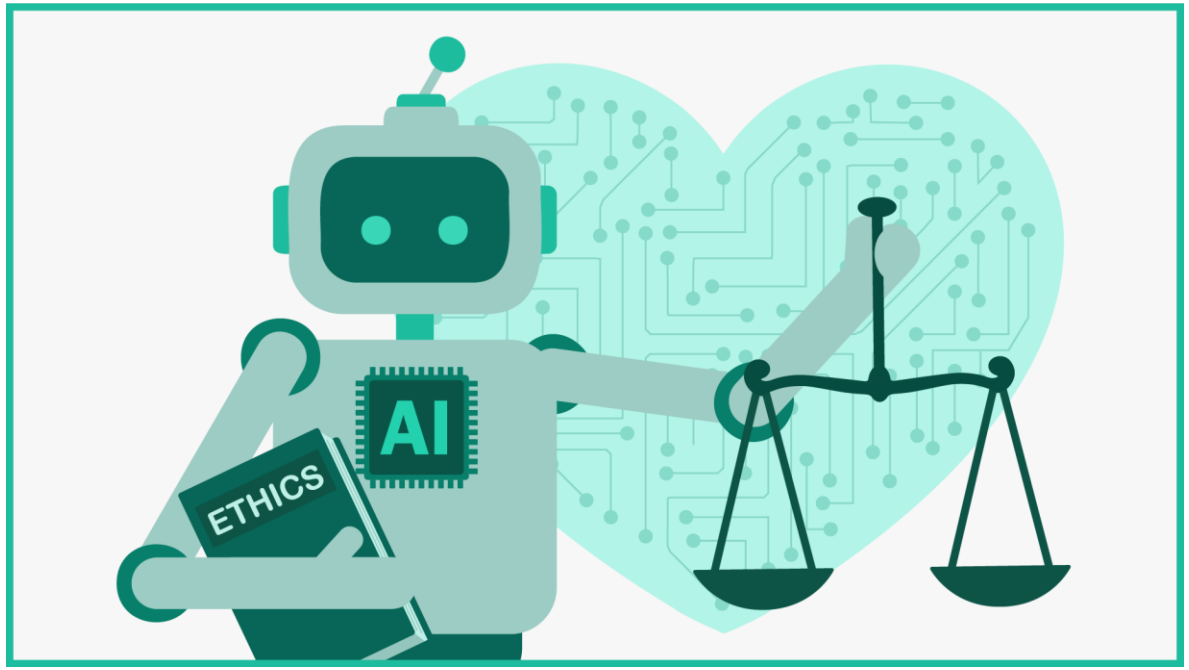
GEB2508 AI and Society

Review of Last Class

- Technology-Environment Relations
  - Environmental Malthusianism vs. Cornucopianism
  - Environmental Cost of AI Infrastructure
  - Environmental Justice and Resource Politics of AI
- AI and Climate Change
  - Tragedy of The Global Carbon Commons
  - Energy Pathways for Powering AI
  - AI for Climate Science

Content

- Theoretical Foundations
  - Classical Ethical Theories
  - Problem of Many Hands
- Ethics of AI
  - Algorithmic Bias & Fairness
  - AI Political Controversies
- Ethics in Practice
  - The Banality of Evil
  - Building Ethical AI

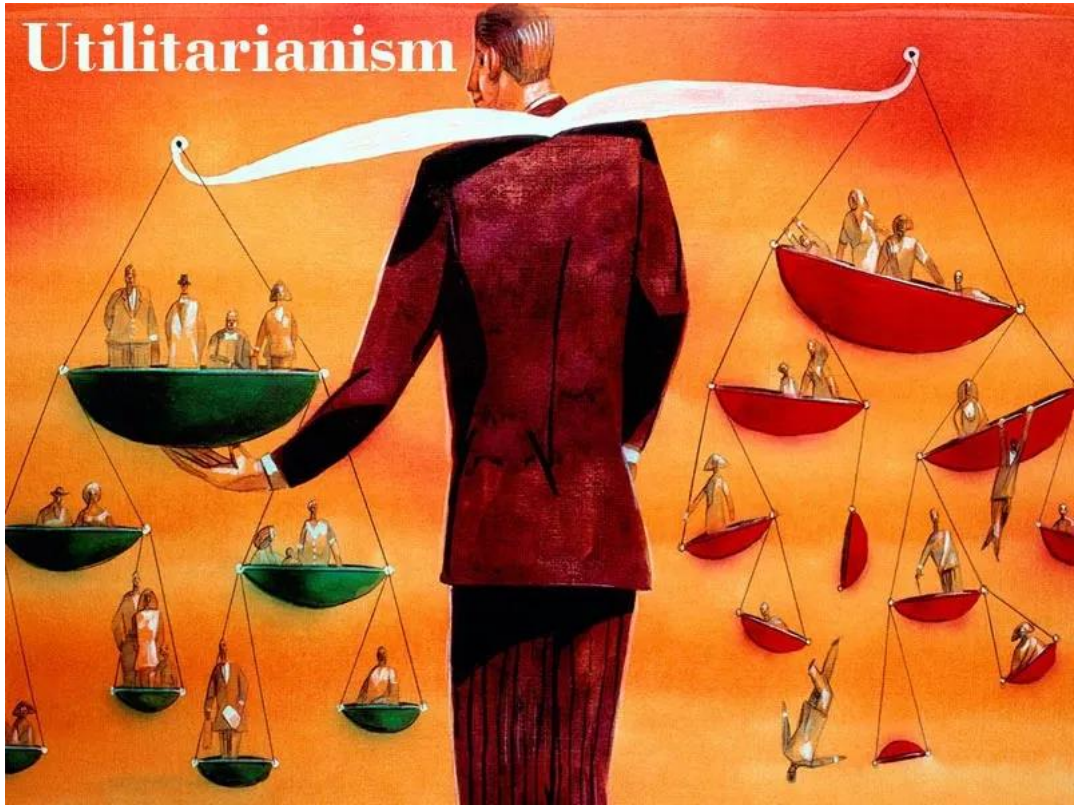


The Ethical Question for AI

- Ethics vs. technical or legal compliance: ethics ≠ law
- Descriptive vs. normative questions: can/is ≠ should
- Individual harm vs. **structural** injustice
- Guiding ethical questions for AI: Who is harmed? Who benefits? Who decides? Who is responsible?

Classical Theories: Utilitarianism

- Key thinkers: Jeremy Bentham, John Stuart Mill
- Focus: **consequences**
- Maximize **overall** welfare or happiness: cost-benefit logic
- AI relevance
  - Aggregated benefits vs. systematic harm to minorities?
  - Surveillance justified by “efficiency”

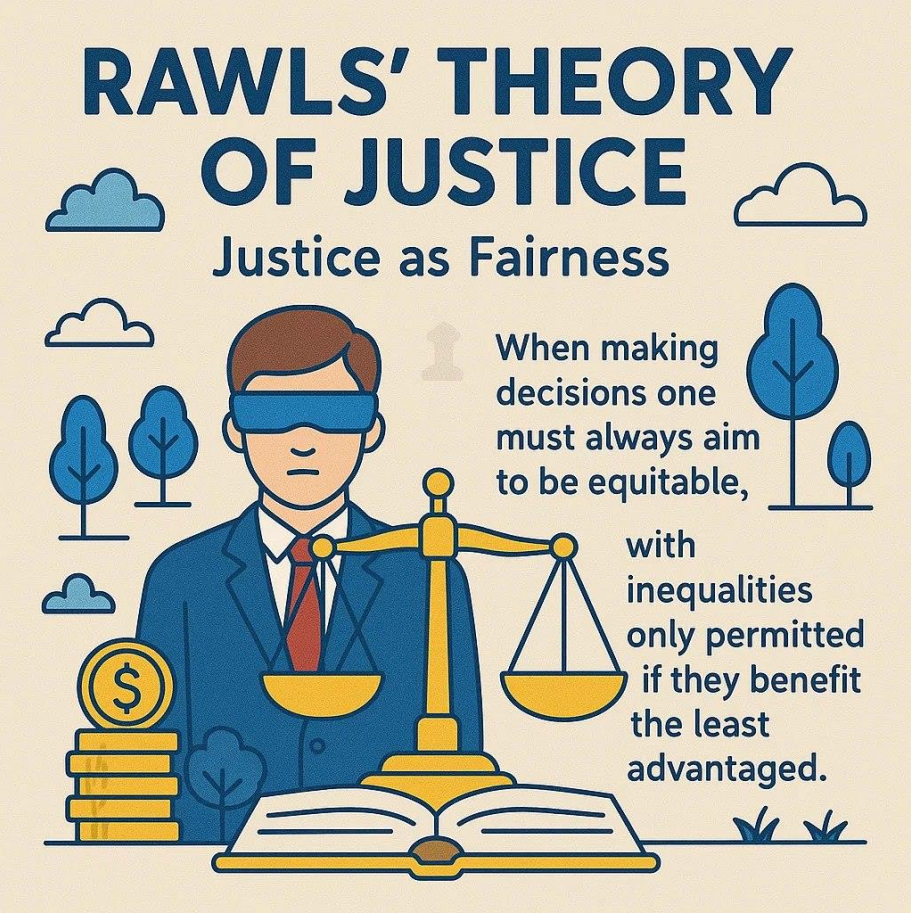


Classical Theories: Deontology

- Key thinkers: Immanuel Kant
- Focus: rights & duties
- People as **ends**, not **means** (e.g., data points)
- Human dignity
- AI relevance
  - **Privacy** as a right (not a trade-off)
  - Informed **consent** for data

Classical Theories: Justice & Fairness

- Key thinkers: John Rawls
- Focus: justice as fairness
- Rules rational people would choose behind a "**veil of ignorance**" → priority to the **least advantaged**
- **Structural** patterns, not isolated cases
- AI relevance
  - Bias as distributive injustice
  - Who does the system systematically disadvantage?

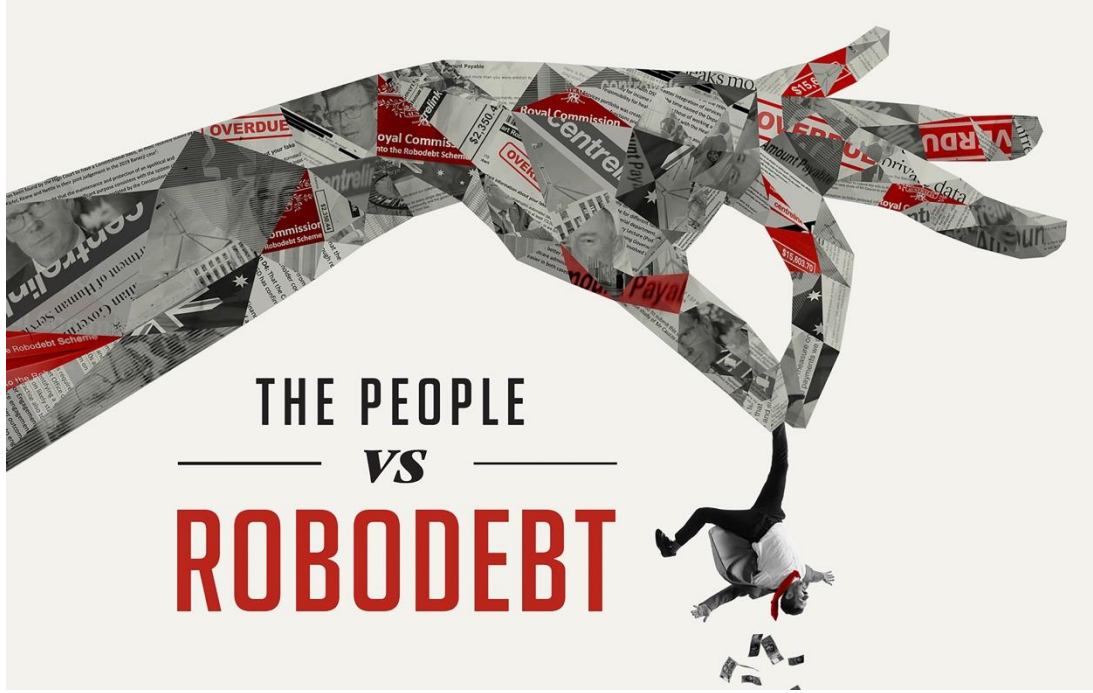


Problem of Many Hands

- AI outcomes are produced by many actors, making moral accountability difficult to assign.
- **Responsibility gap**: harms occur without a clearly accountable actor
- **Black box** narratives and automation rhetoric attribute agency to the algorithm.
- From individual morality to governance & institutions

The Case of Welfare System: Robodebt

- The Australian Government used Robodebt to match tax and welfare data (2016-2019)
- People with casual or seasonal work were wrongly accused of overpayment and forced to disprove debts → **procedural due process** concerns
- dangers of “efficiency-first” automated governance



The Case of Content Moderation: YouTube

- Algorithmic demonetization: creators receive vague or no explanation for demonetization decisions
- Economic governance through algorithms: not formal **censorship**, but **financial** penalties shape what topics are viable
- Disparate impact on sensitive topics & marginalized communities (e.g., war, racism, LGBTQ+ issues)

Algorithmic Bias & Fairness

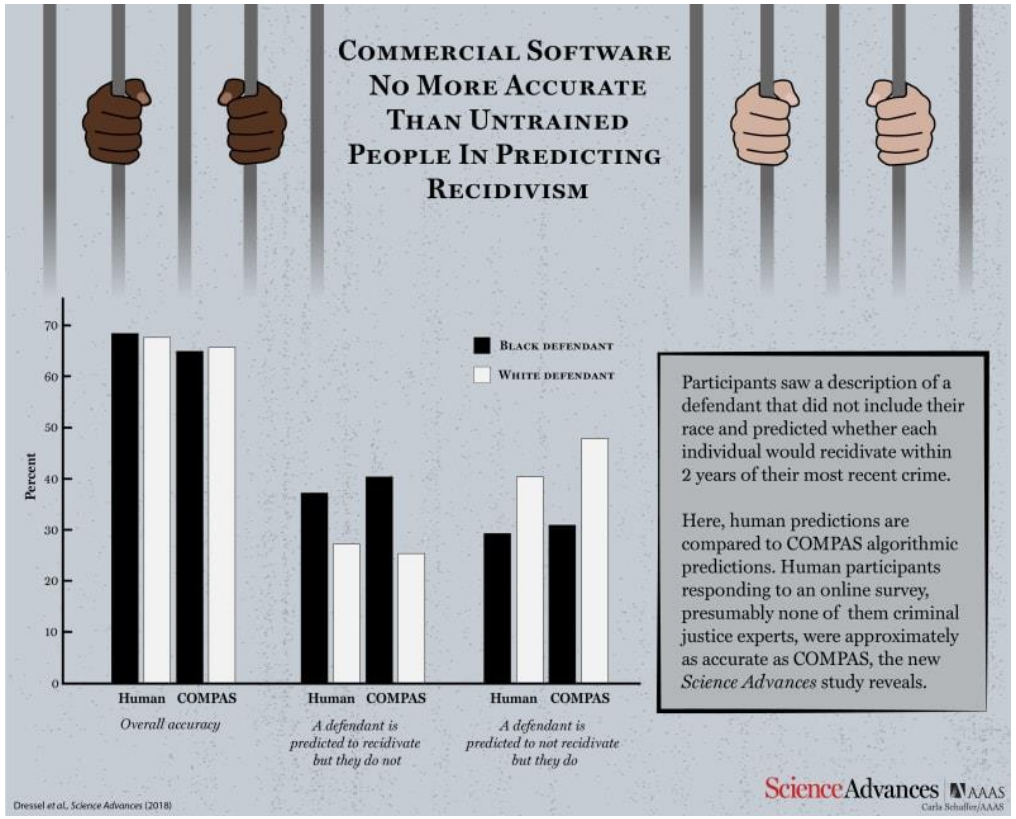
- Bias across the AI lifecycle: data, model design, deployment context
- Historical data as political: accuracy reproduces injustice
- Fairness metrics vs. real-world justice
- Proxy variables (e.g., zip code, shopping behavior, educational background)

The Case of College Admissions: Ofqual

- Context: COVID-19 canceled national exams in UK in 2020 → Ofqual used to standardize teacher-predicted grades
- Heavily weighted schools' historical performance → encoding inequality into the future
- Mass protests → policy reversal → grades replaced by assessments

The Case of Predictive Policing: COMPAS

- COMPAS is an algorithmic risk-scoring tool used in U.S. courts to predict the likelihood of recidivism.
- Black defendants were more likely to be falsely labeled “high risk” than white defendants.
- Fairness as a contested metric
- Biased data and proxies
- Predicting **future** risk vs. evaluating **past** actions



The Case of Credit Scoring: Apple Card

- The Apple Inc. credit card, issued by Goldman Sachs in 2019, was accused of gender discrimination
- Investigation by the New York State Department of Financial Services
- Proxy variables & indirect discrimination
- **Formal** fairness vs. **substantive** fairness

Autonomous Weapons and Delegated Lethality

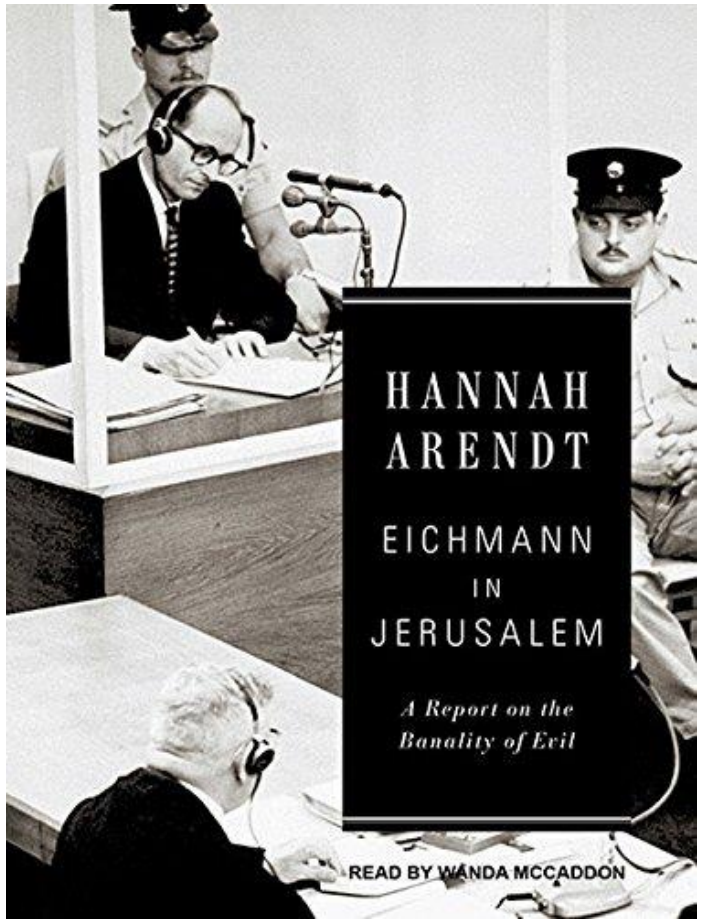
- **Delegation** of life-and-death decisions: should machines be allowed to decide to kill?
- Moral agency and human dignity
- Accountability gaps: who is responsible for war crimes?
- Lowering the threshold for war
- **Distancing** of violence and moral effects

Disinformation and Deepfakes

- Realistic AI-generated synthetic media
- Crisis of epistemic trust
- From harm to individuals to harm to democracy
- AI in elections: microtargeting and voter profiling
- Political communication shifts from an **open**, **public** arena to **personalized** persuasion

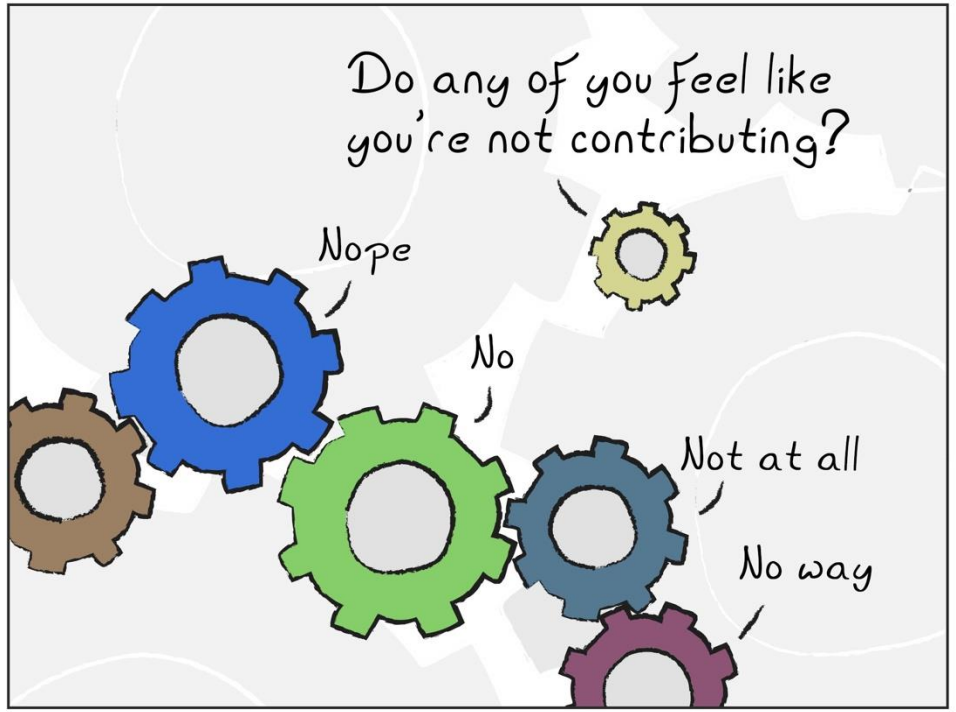
Eichmann in Jerusalem

- Eichmann supervised mass deportations of Jews to extermination camps in Nazi Germany. He was executed in 1962.
- Eichmann was 'neither perverted nor sadistic' but 'terrifyingly normal.'
  - "Eichmann is just you and me."
- Eichmann's defense was that he was '**just following orders**.'
  - Performing evil deeds without evil intentions



The Banality of Evil

- Can one do evil without being evil?
  - Were Germans generally evil, or were the people who committed such acts inherently evil?
  - Was Eichmann an evil sadist or merely a cog in the machine?
  - Why rationality alone isn't enough?
- The **absence of critical thinking** and inability to exercise moral judgment



The Banality of Evil in The Tech Industry

Eichmann's Logic	**Tech Industry Logic**
"I'm just following orders."	"I'm just writing code."
"The decision is above my level."	"Policy decisions aren't my department."
"The system needs me to function."	"If I don't do it, someone else will."
"I'm not pulling the trigger."	"I'm not deploying the weapon."
"I'm just optimizing transport."	"I'm just optimizing targeting algorithms."

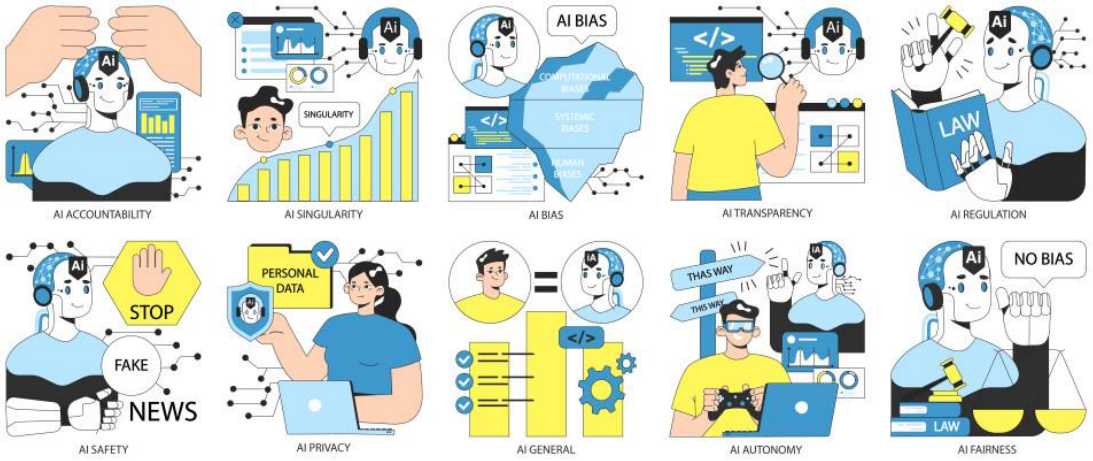
Tech Company Ethics

Google's withdrawal from Project Maven in 2018

Anthropic's restriction of military uses of its systems in 2026

AI Alignment

- Aligning AI systems with human values, intentions, and norms
  - Translating messy human values into machine-optimizable objectives
- Value alignment: pluralism, ambiguity, cultural differences
  - **Whose** values?
- Participatory alignment: public decision-making



Constitutional AI

- Principles: human rights norms; reduce suffering; respect autonomy; honesty and transparency; respect diversity, etc.
- **Self-critique**: AI learns to judge itself
- Ethical resilience through adversarial testing and red teaming
- Teaching **reasoning**, not just rules

Whistleblowing

- Report wrongdoing or illegal activities to the proper authorities
- Refuse to participate in workplace wrongdoing
- Testify in a legal proceeding against the organization
- Leak evidence of wrongdoing to the media

Final Arrangement

Term Paper

- Analyze and compare the responses generated by two different AI models to the same prompt (set).
- DDL: May 8
- Word limit: 1000 words in English

Final Exam

- Time: April 20, 3:30–5:30 PM
- Format: open book, but electronic devices are prohibited.
- Language: students may answer the questions in either Chinese or English.

Sample Exam Questions

I. Definition question (10%×4=40%)
   - Define the term and illustrate it with AI-related examples.
     - Reserve army of labor

II. Reading Question (20%×3=60%)
   - According to Gray (2021), how do digital platforms become geopolitical actors?

Students shall complete all required readings before the exam.

CTE

- Your insights and engagement have enriched our shared journey.
- Your voice matters: help enhance teaching with your feedback.